

Project Template: Compare Multiple LLMs on a Real-World Task

Project Title:	[Compare Multiple LLMs on a Real-World Task.]
Student Name:	[Meghana Sadula]
Cohort / Batch:	[Batch 1]
Date of Submission:	[12-05-2026]
Submitted To:	[Teja Sai Mahesh Chennamasetti]

1. Objective

The goal is to compare multiple LLMs on the same real-world task and evaluate differences in quality, usefulness, reasoning, hallucination risk, and practical applicability.

2. Chosen Real-World Task

Selected Task: [Job Description Matching]

Match a resume with a job description and evaluate how well the candidate fits the role.]

Why this task was chosen: [

Because job seekers need to know if their resume matches a job before applying.]

Real-world scenario/context: [

This is used by job seekers and recruiters to check if a candidate is suitable for a job role and improve chances of selection.]

3. Models Selected

List the specific models you will be evaluating in this project.

Model Name	Provider	Access Type (Free/Paid)
[Qwen 3:1.7b]	[Alibaba Cloud]	[Free]
[Llama 3.2:1b]	[Meta]	[Free]
[phi-3:3.8b]	[Microsoft]	[Free]

4. Prompt Used

Use the same prompt for all models. Do not change wording.

Final Prompt:

[**PROMPT :**

Analyze the following resume against the given job description.

- 1. Give a match score out of 100
- 2. List matching skills
- 3. List missing skills
- 4. Give suggestions to improve the resume
- 5. Do not add any skills that are not present in the resume

Job Description:

[Job Title: AI Engineer]

JOB DESCRIPTION:

We are looking for a skilled AI Engineer to design, develop, and deploy machine learning and AI-based solutions. The ideal candidate should have strong knowledge of data science, machine learning algorithms, and programming skills to build intelligent systems.

Key Responsibilities:

- Develop and deploy machine learning models for real-world applications
- Work with large datasets and perform data preprocessing and analysis
- Build and optimize deep learning models using frameworks like TensorFlow or PyTorch
- Design and implement NLP-based applications
- Collaborate with cross-functional teams including product managers and developers
- Monitor and improve model performance over time
- Integrate AI models into production systems

Required Skills:

- Strong programming skills in Python
- Knowledge of machine learning algorithms (supervised and unsupervised learning)
- Experience with deep learning frameworks (TensorFlow, PyTorch)
- Understanding of Natural Language Processing (NLP)
- Familiarity with data handling libraries like Pandas and NumPy
- Knowledge of APIs and model deployment
- Basic understanding of cloud platforms (AWS, GCP, or Azure)

Preferred Qualifications:

- Experience with LLMs and prompt engineering
- Knowledge of computer vision techniques
- Experience in building real-world AI applications
- Understanding of MLOps concepts
- Bachelor's degree in Computer Science, AI, or related field

Soft Skills:

- Problem-solving ability
- Good communication skills
- Ability to work in a team
- Strong analytical thinking

Experience Level:

- 0-2 years (Entry Level / Internship)

Location:

- Remote / Hybrid

RESUME:

Name: Meghana Sadula

Email: meghana@email.com

Phone: +91-1234567890

LinkedIn: linkedin.com/in/meghana

Location: India

Objective:

Motivated and detail-oriented aspiring AI Engineer with a strong foundation in machine learning, data analysis, and Python programming. Seeking an entry-level role to apply AI skills in real-world applications and contribute to innovative projects.

Education:

Bachelor of Technology in Computer Science

XYZ University, India

2021 - 2025

Technical Skills:

- Programming: Python
 - Machine Learning: Supervised & Unsupervised Learning
 - Deep Learning: Basic Neural Networks, CNN
 - Libraries/Frameworks: TensorFlow, PyTorch, Scikit-learn
 - Data Handling: Pandas, NumPy
 - NLP: Text classification, sentiment analysis
 - Tools: Jupyter Notebook, Git
 - Cloud: Basic knowledge of AWS
-

Projects:

1. Resume Screening System using Machine Learning
 - Built a model to match resumes with job descriptions
 - Used NLP techniques for keyword extraction and similarity scoring
 - Achieved accurate classification using Scikit-learn
 2. Sentiment Analysis on Product Reviews
 - Developed NLP model to classify customer reviews as positive/negative
 - Used Python, Pandas, and Scikit-learn
 - Improved accuracy through preprocessing techniques
 3. Image Classification using CNN
 - Built a convolutional neural network model using TensorFlow
 - Classified images into multiple categories
 - Achieved good accuracy on test dataset
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Internship:

AI/ML Intern (Virtual)
ABC Tech Company

- Assisted in building machine learning models
 - Worked on data preprocessing and model evaluation
 - Collaborated with team on small AI projects
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Certifications:

- Machine Learning Course - Coursera
 - Introduction to AI - Google
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Soft Skills:

- Problem-solving
 - Communication
 - Team collaboration
 - Analytical thinking
-

Achievements:

- Completed multiple AI/ML projects
- Active learner in AI and emerging technologies]

5. Benchmark Check

Conduct background research on your selected models before starting the task evaluation.

Model 1

- **Model Name:** [Qwen 3:1.7b]
- **Provider:** [Alibaba Cloud]
- **Context Window:** [Usually 32K–128K may default to a smaller active context]
- **Speed / Latency:** [Fast — responses are generated quickly with minimal delay]
- **Price:** [Free version available; advanced models and higher usage limits require a paid subscription, for using API we require paid subscription]
- **General Positioning (Reasoning / Fast / Coding / General):** [A balanced open-source AI family optimized for reasoning, coding, multilingual tasks, and practical real-world use
smarter than many small lightweight models
faster than many large reasoning-heavy models]
- **Benchmark / Leaderboard Observation:** [A top-performing open-source AI family with strong benchmark results in reasoning, coding, math, and multilingual tasks while maintaining good efficiency.
]
- **My Real Task Observation:** [Qwen 3:1.7b performed strongly on real-world resume analysis tasks with good reasoning, structured responses, and practical suggestions, though it showed minor instruction-following issues by suggesting skills not present in the resume.]

Model 2

- **Model Name:** [Llama 3.2:1b]
- **Provider:** [Meta]
- **Context Window:** [large (can remember long conversations, supports larger prompts/documents)]
- **Speed / Latency:** [fast (faster than larger 7B/13B models,)]
- **Price:** [Free version available.]

Paid plan for better models and higher usage]

- **General Positioning (Reasoning / Fast / Coding / General):** [A lightweight fast general-purpose open model optimized for local usage and efficient inference.
Not Ideal For: deep reasoning, highly complex planning, advanced agent workflows]
- **Benchmark / Leaderboard Observation:** [Meta Llama 3.2:1B is generally positioned as a fast and efficient lightweight open-source model that delivers solid small-model benchmark performance but is limited in advanced reasoning compared to larger LLMs.]
- **My Real Task Observation:** [Meta Llama 3.2:1B handled the resume-analysis task with good speed and readable output, but its responses were more generic and less contextually intelligent than stronger reasoning-focused models.]

Model 3

- **Model Name:** [phi 3:3.8b]
- **Provider:** [Microsoft]
- **Context Window:** [large supports long conversations, can process large prompts/documents]
- **Speed / Latency:** [Fast]
- **Price:** [Free version available, advanced features in paid plans]
- **General Positioning (Reasoning / Fast / Coding / General):** [A lightweight efficient model optimized for fast local inference and practical small-device usage.]
- **Benchmark / Leaderboard Observation:** [Microsoft Phi-3:3.8B is generally recognized as a highly efficient small language model that delivers strong benchmark performance for its size while prioritizing speed and lightweight deployment over deep reasoning capability.]
- **My Real Task Observation:** [Microsoft Phi-3:3.8B showed decent practical reasoning and fast response generation in the resume-analysis task, but produced more factual inconsistencies and weaker instruction adherence than Qwen and Llama.]

6. Output Comparison

Paste the actual raw output or a detailed summary of the response from each model.

Qwen Response:

1. Match Score: 92/100

Meghana's resume aligns strongly with the job description but has minor gaps and could benefit from refinement.

2. Matching Skills

- **Programming:** Python (core requirement)
- **Machine Learning:** Supervised/unsupervised learning (aligned with job description)
- **Deep Learning:** Basic neural networks, CNN (aligned with frameworks like TensorFlow/PyTorch)
- **NLP:** Text classification, sentiment analysis (aligned with NLP requirements)
- **Data Handling:** Pandas, NumPy (aligned with data preprocessing)
- **Cloud:** Basic AWS knowledge (aligned with job description)
- **Tools:** Jupyter Notebook, Git (aligned with development workflows)
- **Soft Skills:** Problem-solving, communication, teamwork, analytical thinking (aligned with soft skills)

3. Missing Skills

- **LLMs (Large Language Models):** No experience mentioned.
- **Computer Vision:** No relevant projects or skills listed.
- **MLOps:** No experience or knowledge mentioned.
- **Advanced Cloud Experience:** Limited detail (e.g., AWS/Azure/GCP expertise).

4. Suggestions to Improve the Resume

1. Enhance Project Descriptions:

- Add more context to projects (e.g., "Built a real-world NLP model for sentiment analysis with 95% accuracy" instead of "Improved accuracy through preprocessing techniques").
- Highlight specific metrics (e.g., "Achieved 98% accuracy on sentiment analysis" or "Classified 10,000 reviews in 48 hours").

2. Emphasize Internship Experience:

- Detail the internship role (e.g., "Assisted in building a predictive model for customer churn using Python and Scikit-learn") and quantify outcomes (e.g., "Reduced model training time by 30%").

3. Highlight Cloud and MLOps Knowledge:

- Mention AWS/GCP/Azure experience (e.g., "Used AWS S3 for data storage and CloudWatch for monitoring model performance").
- Add a line about MLOps concepts (e.g., "Implemented CI/CD pipelines for model deployment").

4. Add Certifications as Qualifications:

- List Coursera and Google AI courses as "Additional Certifications" to strengthen technical credibility.

5. Refine Soft Skills:

- Use bullet points to showcase soft skills (e.g., "Collaborated with cross-functional teams to deliver AI solutions" or "Led a 3-member team to develop a predictive model").

6. Include Preferred Qualifications:

- Add a note: "Experience with LLMs and prompt engineering (desirable)" to align with job preferences.

5. No Added Skills Not in the Resume

The resume does not include skills like **computer vision**, **LLMs**, **MLOps**, or **advanced cloud platforms** (e.g., GCP/Azure). These are not present in the resume and thus not included in the analysis.

Summary

Meghana's resume is **strongly aligned** with the AI Engineer role, particularly in technical and soft skills. To maximize her chances, she should:

- Add detailed project metrics and outcomes.
- Highlight internship achievements with quantifiable results.
- Emphasize cloud and MLOps experience.
- List certifications as additional qualifications.

Llama 3.2:1b

Match Score: 90 out of 100

Matching Skills:

1. Programming (Python): 8/10
2. Machine Learning: Supervised and Unsupervised Learning, Deep Learning (basic Neural Networks, CNN): 9/10
3. Data Analysis: Pandas, NumPy: 6/10
4. NLP: Text classification, Sentiment analysis: 8/10
5. Cloud Platforms: AWS: 7/10

Missing Skills:

1. Experience with LLMs and Prompt Engineering: Not mentioned in the resume.
2. Knowledge of Computer Vision Techniques: Not mentioned in the resume.
3. Understanding of MLOps Concepts: Not explicitly stated, but implied through experience with data handling libraries and model deployment.

Suggestions to Improve the Resume:

1. Highlight specific projects that demonstrate proficiency in AI/ML techniques, such as LLMs or prompt engineering.
2. Emphasize your understanding of cloud platforms (AWS, GCP, or Azure) and their applications in machine learning.

3. Consider adding relevant certifications or courses related to AI/ML.
4. Quantify achievements by including specific metrics or statistics for each project.
5. Proofread the resume multiple times to ensure there are no grammatical errors.

Additional Tips:

1. Tailor the resume to the specific job description and requirements, highlighting the skills that align with the job title.
2. Use action verbs (e.g., "Developed," "Deployed") to describe your accomplishments in AI/ML projects.
3. Use relevant keywords from the job description to improve search engine visibility.
4. Consider adding a link to a portfolio or GitHub repository showcasing your AI/ML projects.
5. Keep the resume concise and easy to read, with clear headings and bullet points.

phi3:3.8b

Match Score out of 100: I would give a match score of 85 for Meghana's resume against the job description provided, as she has relevant skills but lacks some experience which is critical in this entry-level position that seeks candidates with strong knowledge and hands-on projects.

- Matching Skills:
 - Programming (specifically Python)
 - Machine Learning (supervised & unsupervised learning techniques mentioned explicitly). However, she does not state her experience level clearly in the resume which should ideally be at least one year as per job description requirements. It's good that Meghana has some exposure to NLP and CNN through projects but there is no mention of specific algorithms or frameworks used for designing DL models using TensorFlow/PyTorch, nor experience with APIs or model deployment mentioned in her resume.
- Missing Skills:
 - Experience level explicitly stated (it should be at least one year).
 - Familiarity with cloud platforms beyond basic knowledge of AWS as specified by the job description preferred qualifications are more thorough understanding and experience on GCP or Azure in addition to AWS.
- Suggestions for Improvement:
 - Meghana could clearly state her previous internship's duration (if any) related to AI/ML projects which would help improve the match score, if not already present under 'Experience'.
 - She should add specific details about frameworks and libraries used in deploying machine learning models. If she has worked with APIs or

model deployment during her internship at ABC Tech Company (if any experience), it needs to be mentioned for better alignment with the job description requirements, such as TensorFlow/PyTorch specifically instead of generic 'Basic knowledge'.

- She could emphasize more on relevant projects and achievements in machine learning or AI where she applied her skills directly. For example: "Developed a resume screening system using supervised NLP techniques that leveraged TensorFlow to score the similarity between job descriptions and applicants' CVs, achieved with an accuracy rate of over X%." This would illustrate practical application aligning closely with key responsibilities.

Model 1 Output- Qwen

["Qwen 3:1.7b — Best overall for structured, realistic, and recruiter-style resume evaluation."]

Model 2 Output- Llama

["Llama 3.2:1b — Best for concise ATS-focused feedback with simple readability."]

Model 3 Output- phi

["Phi-3:3.8b — Better for broad reasoning attempts, but weaker in factual precision and instruction adherence."]

7. Analysis

Answer the following questions based on your observations of the model outputs.

Which model gave the clearest response? [Qwen 3:1.7b Most structured, readable and professionally organized.]

Which model followed the prompt most accurately? [Llama 3.2:1b Stayed closer to prompt constraints with fewer unsupported additions.]

Which model was most useful in a real-world scenario? [Qwen 3:1.7b Provided the most actionable and recruiter-like feedback]

Which model showed hallucination or overconfidence? [Phi-3:3.8b Added inaccurate assumptions and incorrect interpretations]

Which model handled ambiguity better? [Qwen 3:1.7b Better contextual understanding and interpretation of resume-JD relationships.]

Key strengths of each model:

[Qwen 3:1.7b→ Resume analysis, Structured evaluation tasks, Real-world practical feedback

Llama 3.2:1b → Quick analysis, Lightweight local usage, Basic resume screening tasks

Phi-3:3.8b → Lightweight experimentation, Low-resource environments, General small-model tasks]

Key weaknesses of each model:

[Qwen 3:1.7b→ Minor instruction violations, Suggested skills not present in resume, Slight tendency toward over-enhancement suggestions.

Llama 3.2:1b → Generic/template-style feedback, Less deep reasoning, Weaker contextual understanding, Limited personalization.

Phi-3:3.8b → Most hallucinations/inaccuracies, Misinterpreted job requirements, Lower factual precision, Weak instruction adherence, Overconfident assumptions]

8. Final Verdict

Best Performing Model: [Qwen]

Weakest Performing Model: [phi]

Did benchmark results match real-world output? [

The experiment showed that benchmarks are useful indicators of model capability, but real-world prompt handling, instruction following, and contextual reasoning can differ significantly during practical tasks.]

Final Recommendation (with reasoning): [Among the tested models, Qwen 3:1.7b provided the best combination of reasoning, structure, and practical usefulness, making it the strongest choice for real-world resume-analysis tasks.]

9. Submission Checklist

- ✓ Selected only one task
- ✓ Compared at least 3 models
- ✓ Used the same prompt for all models
- ✓ Completed benchmark check
- ✓ Compared outputs properly
- ✓ Wrote final conclusion

Focus on real output quality and usefulness, not just model popularity or answer length.